Probabilistic Combinatorics

Sheet 5

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Problem 1. Show that for the Erdős-Renyi model, if $p \ll 1/n$, then

$$\mathbb{P}(G(n, p_n) \text{ contains a cycle}) \to 0$$

Here, a cycle (of length k) in a graph $G = (V, E)$ is a sequence of k pairwise distinct vertices $v_1, \ldots, v_k \in V$, such that $\{v_i, v_{i+1}\}$ for $1 \leq i < k$ and $\{v_k, v_1\}$.

Proof.
Recall that the Erdős-Renyi model $G(n, p_n)$ is the discrete probability space $(\Omega_n, \mathbb{P})$ where the sample space Ω_n is the set of all undirected graphs $G = ([n], E)$ while the probability measure $\mathbb{P}(G) = p_n^{|E|}(1 - p_n)^{n-|E|}$ for any graph $G \in \Omega_n$.

Next, we define the random variable $X_n : \Omega_n \to \mathbb{N}$ that assigns to each graph $G \in \Omega_n$ the number of cycles in G, and the original problem turns into the problem of showing that

$$\lim_{n\to\infty} \mathbb{P}(X_n > 0) = 0$$

Based on the definition of cycle given above, any cycle in G must involve at least 3 distinct vertices in G. That is, for any integer $k \in [3, n]$, we define the random variable $X_n^{(k)} : \Omega_n \to \mathbb{N}$ that assigns to each graph $G \in \Omega_n$ the number of cycles of length k in G. Moreover, it is obvious that $X_n = \sum_{k=3}^{n} X_n^{(k)}$

Let S_k be the set of all possible sequences of k distinct vertices in G. Fixed a sequence $S \in S_k$, let the indicator random variable $X_n^{(S)} : \Omega_n \to \mathbb{N}$ defined by the following rule for any graph $G \in \Omega_n$:

$$X_n^{(S)}(G) = \begin{cases} 1 \text{ if } S \text{ is a cycle in } G \\ 0 \text{ otherwise} \end{cases}$$

This implies that $X^{(k)} = \sum_{S \in S_k} X_n^{(S)}(G)$. Next, we define the indicator variable $X_n^{(S)} = 1_{A_n^{(S)}}$ of the event

$$A_n^{(S)} = \{G = ([n], E) | S \text{ is a cycle in } G\}$$

then $\mathbb{E}(X_n^{(S)}) = \mathbb{E}(1_{A_n^{(S)}}) = \mathbb{P}(A_n^{(S)})$. It is known that, for an arbitrary graph $G \in \Omega_n$, the sequence of vertices $S = v_1 v_2 \ldots v_k$ is a cycle when the edges $\{v_i, v_{i+1}\}$ and $\{v_k, v_1\}$ are presented in G for $1 \leq i < k$. That is, such k edges must be presented in G so that S can be a cycle in G. Since the probability of choosing a graph $G \in \Omega_n$ having a specific edge is p_n. Hence,

$$\mathbb{P}(A_n^{(S)}) = p_n^k \tag{1.1}$$

Recall that $X_n^{(k)} = \sum_{S \in S_k} X_n^{(S)}$. By linearity of expectation, we have

$$\mathbb{E}(X_n^{(k)}) = \sum_{S \in S_k} \mathbb{E}(X_n^{(S)}) = \sum_{S \in S_k} \mathbb{P}(A_n^{(S)}) \tag{1.2}$$

As $\mathbb{P}(A_n^{(S)})$ is known due to the equation (1.1), we now want to calculate $|S_k|$ in order to calculate $\mathbb{E}(X_n^{(k)})$ based on the equation (1.2). In order to construct S_k, we firstly select k distinct vertices from G. For each selection, there are $k!$ ways to form different sequences. Moreover, there are $\binom{n}{k}$ ways to choose k distinct vertices from G; hence $|S_k| = \binom{n}{k}k!$. Therefore, by the equation (1.2), we have

$$\begin{aligned}\mathbb{E}(X_n^{(k)}) &= \binom{n}{k}k!p_n^k \\ &= \frac{n(n-1)\dots(n-k+1)(n-k)!}{k!(n-k)!}k!p_n^k \\ &= n(n-1)\dots(n-k+1)p_n^k \\ &\leq (np_n)^k \end{aligned} \tag{1.3}$$

Due to the equation (1.3) and the fact that $X_n = \sum_{k=3}^{n} X_n^{(k)}$, by linearity of expectation, we have,

$$\mathbb{E}(X_n) = \sum_{k=3}^{n}\mathbb{E}(X_n^{(k)}) \leq \sum_{k=3}^{n}(np_n)^k \tag{1.4}$$

Let $a = np$, and thus $\sum_{k=3}^{n}(np^n)^k = \sum_{k=3}^{n}a^k$. It is known from the sum of geometric series that $\sum_{k=0}^{n}a^k = \frac{a^{n+1}-1}{a-1}$. Therefore,

$$\begin{aligned}\sum_{k=3}^{n}(np_n)^k &= \sum_{k=3}^{n}a^k \\ &= \frac{a^{n+1}-1}{a-1} - a^2 - a^1 - a^0 \\ &= \frac{a^{n+1}-a^3}{a-1}\end{aligned} \tag{1.5}$$

Moreover, since $p_n \ll 1/n$, then $p_n = o(1/n)$. This implies that $\lim_{n\to\infty} np = 0$, or $\lim_{n\to\infty} a = 0$. This implies that $\lim_{n\to\infty}\frac{a^{n+1}-a^3}{a-1} = 0$. By the equation (1.5), we have $\lim_{n\to\infty}\sum_{k=3}^{n}(np_n)^k = 0$. It follows from the equation (1.4) that

$$\lim_{n\to\infty}\mathbb{E}(X_n) = 0$$

this implies that $\mathbb{E}(X_n) = o(1)$. It follows from the Corollary 6.3 in the lecture that

$$\lim_{n\to\infty}\mathbb{P}(X_n > 0) = 0$$

and the proof is finished. □

Problem 2. Show that

1) For $n \geq 3$ and a graph $G = (V, E)$ with $\#V = n$ and $\#E \geq n$, there is a cycle in G.
2) $p(n) = 1/n$ is a threshold function for the monotone graph property "$G(n, p)$ has a cycle".

Proof.
The statement (i) can be proved using induction on n. That is, we firstly show that such statement holds for the base case where $n = 3$. In this case, since there are at least 3 edges in G, then there must be an edge between any two of 3 vertices in G, and thus G contains at least one cycle. Therefore, the statement holds for the base case.

Next, suppose inductively that G contains a cycle when $|V| = n$ and $|E| \geq n$ for $n \geq 3$, we want to show that G also contains a cycle when $|V| = n + 1$ and $|E| \geq n + 1$. In order to do so, we consider the following cases

- There is at least one vertex in G having degree of either 0 or 1. In this case, fixed a vertex v in G having such property, then the induced subgraph $G^{'}$ of G, where its vertices are in the set $V - \{v\}$, has n vertices and either $n+1$ or n edges when degree of v is either 0 or 1 respectively. It follows from the induction hypothesis that $G^{'}$ contains a cycle, and thus G contains a cycle since $G^{'}$ is a subgraph of G.
- Otherwise, any vertex in G has degree of at least 2. Let $P = \{v_1, v_2\}, \{v_2, v_3\}, \ldots, \{v_{k-1}, v_k\}$ be the longest path in G. Since the vertex v_k has degree of at least 2, then there must be an edge between v_k and any other vertex v in G that is not v_{k-1}. If such vertex does not belong to P, then there exists a path in G that is longer than P. However, this is a contradiction to the fact that P is the longest path in G. This implies that v must belong to P. Therefore, v_k must have an edge to v_i where $1 \leq i \leq k-2$. Therefore, the sequence of vertices $v_i, v_{i+1}, \ldots, v_k$ is a cycle in G.

Due to the reasons given above, a graph G with $|V| = n+1$ and $|E| \geq n+1$ must contain a cycle when another graph G with $|V| = n$ and $|E| \geq n$ contains a cycle. Therefore, we have closed the induction, and the proof is finished for the statement (i).

Regarding the statement (ii), due to the Problem 1, it remains to show that when $p_n \gg 1/n$, then

$$\mathbb{P}(G(n, p_n) \text{ contains a cycle}) \to 1$$

Based on the statement (i), it is known that any graph with n vertices having at least n edges must contain a cycle. That is, the idea to prove this is to show that the number of edges in any graph that is generated by $G(n, p_n)$ must be at least n when n goes to infinity.

Recall that the Erdős-Renyi model $G(n, p_n)$ is the discrete probability space $(\Omega_n, \mathbb{P})$ where the sample space Ω_n is the set of all undirected graphs $G = ([n], E)$ while the probability measure $\mathbb{P}(G) = p_n^{|E|}(1-p_n)^{n-|E|}$ for any graph $G \in \Omega_n$.

Due to the idea described above, we define the random variable $X_n : \Omega_n \to \mathbb{N}$ that assigns to each graph $G \in \Omega_n$ the number of edges in G. Let S be the set of all unordered pairs of vertices in G. Fixed a pair $s \in S$, let $X_n^{(s)} : \Omega_n \to \mathbb{N}$ be the indicator random variable defined by the following rule for any graph $G \in \Omega_n$:

$$X_n^{(s)} = \begin{cases} 1 \text{ if the edge } s \text{ is presented in } G \\ 0 \text{ otherwise} \end{cases}$$

Since the probabilities to choose a graph $G \in \Omega_n$ having and not having a specific edge are p_n and $1-p_n$ respectively, then $\mathbb{P}(X_n^{(s)} = 1) = p_n$ and $\mathbb{P}(X_n^{(s)} = 0) = 1 - p_n$. Moreover, since $X_n = \sum_{s \in S} X_n^{(s)}$, then X_n must have binomial distribution. This is because $X_n^{(s)}$ is an indicator random variable and the fact that $X_n^{(s_1)}$ and $X_n^{(s_2)}$ are independent from each other for distinct $s_1, s_2 \in S$. Therefore, we have $\mathbb{E}(X_n) = |S|p_n$ and $\text{Var}(X_n) = |S|p_n(1-p_n)$. Since any graph $G \in \Omega_n$ is undirected, then $|S| = \binom{n}{2}$, and thus

$$\mathbb{E}(X_n) = \binom{n}{2} p_n \text{ and } \text{Var}(X_n) = \binom{n}{2} p_n(1-p_n)$$

... □

Problem 3. Show for a monotone graph $\mathcal{P} = \{\mathcal{P}_n\}_{n \in \mathbb{N}}$ and parameters $0 \leq p_1 \leq p_2 \leq 1$ that

$$\mathbb{P}(G(n, p_1) \in \mathcal{P}) \leq \mathbb{P}(G(n, p_2) \in \mathcal{P}).$$

Here $\mathbb{P}$ denotes the probability induced the corresponding parameter in $G(n, p)$

Proof.
Let p be any real number in the interval $[0, 1]$, we define $p_2 = p_1 + (1-p_1)p$. Next, let G and G_1 be graphs that are randomly generated by $G(n, p)$ and $G(n, p_1)$, respectively.

Now, we define $G_2 = G \cup G_1$, this means that G_2 has n vertices and the edges of G_2 are in $E(G) \cup E(G_1)$. Moreover, this implies that an edge in G_2 can only exist when it is presented in either G or G_1. Recall that in $G(n, p)$, the probability to generate a graph having and not having a specific edge is p and $1 - p$, while in $G(n, p_1)$ they are p_1 and $1 - p_1$. That is, the probability of choosing graphs G and G_1 in $G(n, p)$ and $G(n, p_1)$ respectively, such that G_2 can have a specific edge e is actually the sum of the probabilities of the following events

1. Edge e is presented in G_1, but not in G. The probability of this event is $p_1(1 - p)$
2. Edge e is presented in G, but not in G_1. The probability of this event is $p(1 - p_1)$
3. Edge e is presented in both G and G_1. The probability of this event is $p_1 p$

Therefore, the probability that $G(n, p)$ and $G(n, p_1)$ generate G and G_1 respectively, such that a specific edge e is presented in G_2 is

$$p_1(1 - p) + (1 - p_1)p + p_1 p = p_1 + (1 - p_1)p$$

Combine this fact with our definition that $p_2 = p_1 + (1 - p_1)$, we can conclude that G_2 is a graph that is randomly generated by $G(n, p_2)$.

Recall that $G_2 = G \cup G_1$. In fact, this union can be perceived as an addition of edges, which are presented in a graph G, to a graph G_1, and the result of such addition is a graph G_2. It follows that $G(n, p_2) \in \mathcal{P}$, since $\mathcal{P}$ is a monotone graph property and $G(n, p_1) \in \mathcal{P}$. That is, it remains to show that $\mathbb{P}(G(n, p_1) \in \mathcal{P}) \leq \mathbb{P}(G(n, p_2) \in \mathcal{P})$. Since $G_2 = G \cup G_1$, then G_1 is a subgraph of G_2, and thus $\mathbb{P}(G(n, p_1) \in \mathcal{P}) \leq \mathbb{P}(G(n, p_2) \in \mathcal{P})$ as $\mathcal{P}$ is a monotone graph property and the number of edges in G_2 is always more than or equal to the number of edges in G_1. Hence, the proof is finished. □